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Subject oop
lab task

QUESTION N0 1:

Vehicle Inheritance

code:

```
class Vehicle { // name babar Zareen Sap id 74760
    String make, model, fuelType;
    int year;

    Vehicle(String make, String model, int year, String fuelType) {
        this.make = make;
        this.model = model;
        this.year = year;
        this.fuelType = fuelType;
    }

    void fuelEfficiency() {
        System.out.println("Calculating fuel efficiency...");
    }

    void distanceTravelled() {
        System.out.println("Calculating distance travelled...");
    }

    void maxSpeed() {
        System.out.println("Calculating max speed...");
    }
}
```

```
// Truck class     //name babar Zareen Sap id 74760
class Truck extends Vehicle {

    Truck(String make, String model, int year, String fuelType) {
        super(make, model, year, fuelType);
    }

    void maxSpeed() {
        System.out.println("Truck max speed: 120 km/h");
    }
}

// Car class      //name babar Zareen Sap id 74760
class Car extends Vehicle {

    Car(String make, String model, int year, String fuelType) {
        super(make, model, year, fuelType);
    }

    void maxSpeed() {
        System.out.println("Car max speed: 180 km/h");
    }
}
```

```java
// Motorcycle class    //name babar Zareen Sap id 74760
class Motorcycle extends Vehicle {

    Motorcycle(String make, String model, int year, String fuelType) {
        super(make, model, year, fuelType);
    }

    void maxSpeed() {
        System.out.println("Motorcycle max speed: 150 km/h");
    }
}

// Main class
public class Main {
    public static void main(String[] args) {

        Car c = new Car("Toyota", "Corolla", 2022, "Petrol");
        Truck t = new Truck("Isuzu", "TruckX", 2020, "Diesel");
        Motorcycle m = new Motorcycle("Honda", "CD70", 2021, "Petrol");

        c.fuelEfficiency();
        c.distanceTravelled();
        c.maxSpeed();
```

```java
        System.out.println();
                                    //name babar Zareen Sap id 74760
        t.fuelEfficiency();
        t.distanceTravelled();
        t.maxSpeed();

        System.out.println();

        m.fuelEfficiency();
        m.distanceTravelled();
        m.maxSpeed();
    }
}
```

output:

```
Calculating fuel efficiency...
Calculating distance travelled...
Car max speed: 180 km/h

Calculating fuel efficiency...
Calculating distance travelled...
Truck max speed: 120 km/h

Calculating fuel efficiency...
Calculating distance travelled...
Motorcycle max speed: 150 km/h


...Program finished with exit code 0
Press ENTER to exit console.
```

QUESTION N0 2:

Overloading Compile-time Polymorphism

code:

```
class vehicle   //Name babar zareen Sap id 74760
{
    void run(int a)
    {
        System.out.println("Running");
    }
    int run()           //Name babar zareen Sap id 74760
    {
        System.out.println("car is Running");
        return 0;
    }
    public static void main(String args[])
    {
        vehicle v = new vehicle();
        v.run(5); // call first method
        v.run(); // call second method
    }
}
```

QUESTION N0 3:

Overriding Runtime Polymorphism

code:

```
class vehicle   //Name babar zareen Sap id 74760
{
    void run()
    {
        System.out.println("Running");
    }
}
    class bike extends vehicle
  {
    void run()          //Name babar zareen Sap id 74760
    {
        System.out.println("car is Running");

    }
  }
  class exe
  {
    public static void main(String args[])
    {
        bike b = new bike();
        b.run();
    }
}
```

QUESTION N0 4:

Polymorphism + Inheritance:

code:

```java
class Shape {       //Name babar zareen Sap id 74760

    void draw() {
        System.out.println("Drawing shape...");
    }

    double calculateArea() {
        return 0;
    }
}

// Circle class
class Circle extends Shape {    //Name babar zareen Sap id 74760
    double radius = 5;

    void draw() {
        System.out.println("Drawing Circle");
    }

    double calculateArea() {
        return Math.PI * radius * radius;
    }
}
```

```java
// Cylinder class
class Cylinder extends Shape {
    double radius = 5;
    double height = 10;

    void draw() {       //Name babar zareen Sap id 74760
        System.out.println("Drawing Cylinder");
    }

    double calculateArea() {
        return 2 * Math.PI * radius * (radius + height);
    }
}

// Main class
public class Main {     //Name babar zareen Sap id 74760
    public static void main(String[] args) {

        Shape s;

        s = new Circle();
        s.draw();
        System.out.println("Circle Area: " + s.calculateArea());
```

```java
        System.out.println();

        s = new Cylinder();
        s.draw();
        System.out.println("Cylinder Surface Area: " + s.calculateArea());
    }
}
```

output:

```
Drawing Circle
Circle Area: 78.53981633974483

Drawing Cylinder
Cylinder Surface Area: 471.23889803846896


...Program finished with exit code 0
Press ENTER to exit console.
```